

CS2430 MOBILE COMPUTING

Exercise 2: Memory Matching Flutter Game

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1. Project Overview

This project implements a memory matching mobile game. The player taps cards to reveal animal images, tries to find matching pairs, and wins after all pairs are matched. The app uses a simple in-memory list of card objects and does not require a database.

2. Implemented Requirements

Requirement	Implementation
Use Flutter/Dart	The complete app is written in Dart in lib/main.dart using Flutter Material widgets.
Memory matching gameplay	The app has 16 cards, 8 pairs, card flipping, matching, mismatch delay, and win detection.
Screenshots	Screenshots are included in the screenshots folder and shown below.
Images	Animal image assets and a card back asset are included in assets/images.
ZIP package	This folder can be zipped and uploaded to D2L or GitHub.

3. How to Configure and Run

1. Install Flutter SDK from flutter.dev.
2. Install Android Studio and enable the Flutter and Dart plugins.
3. Open the MemoryMatchFlutter folder in Android Studio.
4. Open the terminal and run: flutter pub get.
5. Start an Android emulator or connect a real Android device.
6. Run the app using Android Studio Run button or the command: flutter run.

4. Exact Project Structure

```
MemoryMatchFlutter/  
├── pubspec.yaml  
├── README.md  
├── README.pdf  
├── lib/  
│   └── main.dart  
├── assets/  
│   └── images/
```

```
|      |—— animal_cat.png
|      |—— animal_dog.png
|      |—— animal_fox.png
|      |—— animal_koala.png
|      |—— animal_lion.png
|      |—— animal_panda.png
|      |—— animal_rabbit.png
|      |—— animal_zebra.png
|      |—— card_back.png
|—— screenshots/
|      |—— game_start.png
|      |—— game_play.png
|      |—— game_win.png
|—— android/
|      |—— app/
|          |—— src/
|              |—— main/
|                  |—— AndroidManifest.xml
```

5. Screenshots

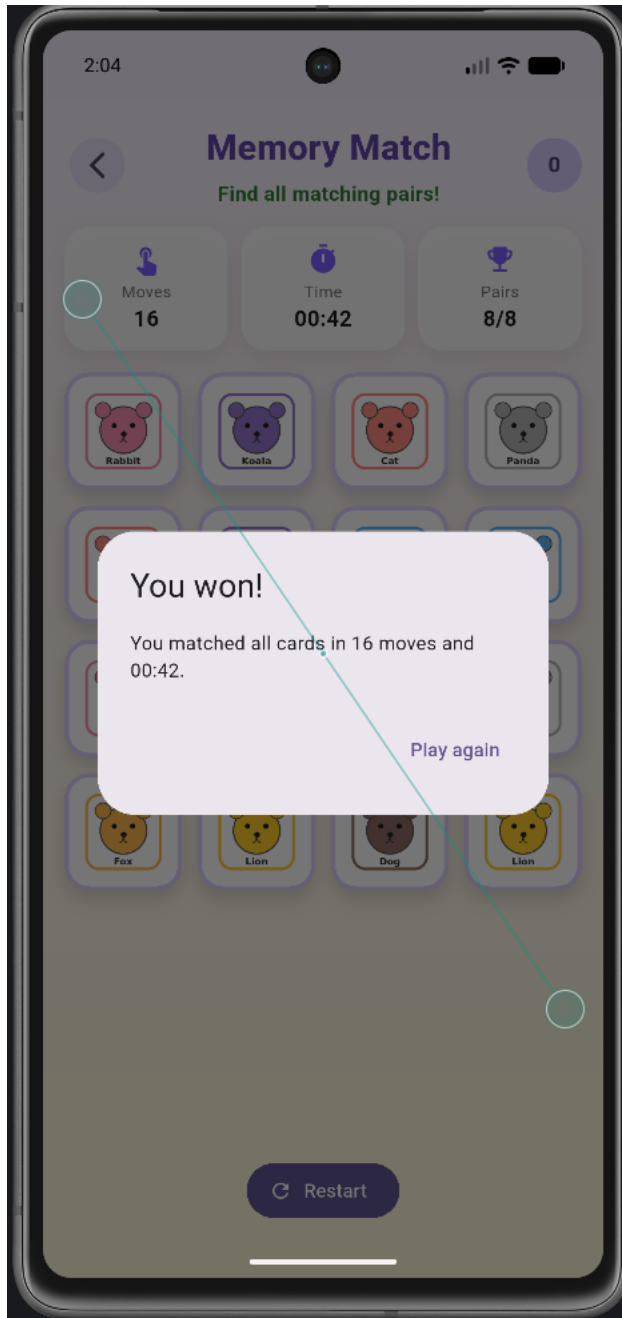
Start



Process



Winning



6. Important Files

lib/main.dart contains the game logic, including the card model, shuffled deck, tap handling, matching logic, timer, counters, restart button, and win dialog. pubspec.yaml registers the assets/images folder so the card

images load correctly.

GitHub and submission

Create a public GitHub repository,

<https://github.com/yueren257-282931/Excecise-1-YueRen1269195.git>

I upload the complete *MemoryMatchFlutter* project folder, then zip the same folder and upload it to D2L. Include all files, generated screenshots, image resources, and this README.pdf.